

Joseph Kerry

West Bridgford, Nottingham • [GitHub](#) • [portfolio](#)

Employment

CoreFlow Operations Subcontractor (Feb 2026 – May 2026)

Part-Time Systems Developer/ Administrator

- Consulting documentation and requirements for SMEs.
- Developing systems from a user perspective to ensure usability and functionality.
- Partial development of Nottingham Forest Supporters Trust website.

Education

Master Of Science at NOTTINGHAM TRENT UNIVERSITY

MSc, Software Engineering

Focus: C#, Java, Python

Major Project: Benchmarking CNN, Hybrid and Transformer Architectures for EEG Motor Imagery Classification on Open BCI's Galea Headset using Standardised Preprocessing Pipelines.

Average Grade: Low Distinction

Bachelor Of Law with Honours at NOTTINGHAM TRENT UNIVERSITY

LLB Hons, Law with Business

Final Grade: Second Class Honours

Notable Projects

Convolutional Vision Transformer – Temporal Convolutional Network Hybrid Model

- Implemented a CVT-TCN using PyTorch for motor imagery classification (not for major project) on the EEGMMIDB dataset converting raw time trials into time-frequency representations.
- Achieved a subject independent accuracy on 4 class motor imagery, achieving similar results to EEGNet with an end-to-end pipeline.

Distributed Cloud Load Balancer

- Built a Java based distributed cloud storage simulator across four dockerised server nodes, implemented client-side AES encryption, file chunking and load balanced distribution.
- Designed a dual database architecture with an MQTT based host manager supporting dynamic scaling.

High Energy Physics Data Pipeline

- Python and FastAPI based datapipeline and analysis model for processing high-energy physics event data. Implemented async ingestion methods, event filtering and statistical analysis endpoints.
- Designed a scalable REST API architecture with optimized query handling and structured data storage to handle high output datasets.

Cover Letter Generator

- Built a full-stack web application that accepts a user CV and a target job description and generates a tailored cover letter in 10 seconds, with dynamic tone adjustment.
- Collaborated with another independent developer using git and Postman.

Skills & Interests

Languages: Python, C#, C, Java

Technical skills: Docker, Postman, React, Node, PyTorch, API handling

Databases: SQL, SQLite, PostgreSQL, Firebase